

CUSTOMER SHOPPING BEHAVIOUR ANALYSIS

End-to-End Data Analytics Project
Python | PostgreSQL | SQL | Power BI

Executive Summary

This project analyzes customer shopping behavior using a dataset of 3,900 retail transactions. The objective was to identify purchasing trends, customer segments, product performance, and revenue drivers. Python was used for data cleaning and transformation, PostgreSQL for data storage, SQL for business analysis, and Power BI for dashboard development. The analysis provides insights that can help improve customer engagement, optimize marketing strategies, and support data-driven decision-making.

Business Problem

A retail company wants to understand customer shopping behavior to improve sales performance, customer satisfaction, and long-term customer loyalty. The analysis focuses on identifying spending patterns, customer demographics, subscription behavior, product performance, and purchase trends.

Project Objectives

- Analyze customer purchasing patterns.
- Identify high-value customer segments.
- Evaluate the impact of subscriptions and discounts.
- Discover top-performing product categories.
- Build an interactive dashboard for business stakeholders.
- Provide actionable business recommendations.

Dataset Overview

Dataset Name: Customer Shopping Trends Dataset

Total Records: 3,900

Total Features: 18

Data Includes: Customer demographics, purchase details, review ratings, subscription status, shipping preferences, discounts, payment methods, and purchase frequency.

Methodology

Data Collection → Data Cleaning (Python) → PostgreSQL Database → SQL Analysis → Power BI Dashboard → Business Insights & Recommendations

Data Preparation & Cleaning

Key preprocessing activities performed:

- Loaded dataset into Pandas DataFrame.
- Explored dataset using `info()` and `describe()`.
- Handled missing values in `review_rating`.
- Standardized column names using `snake_case` format.
- Created `age_group` feature.
- Created `purchase_frequency_days` feature.
- Removed redundant `promo_code_used` field.
- Loaded cleaned dataset into PostgreSQL table 'customer'.

SQL Business Analysis

Business questions answered:

1. Revenue generated by male vs female customers.
2. Customers using discounts while spending above average.
3. Top 5 products by average review rating.
4. Comparison of Standard and Express shipping.
5. Subscriber vs Non-Subscriber spending analysis.
6. Products with highest discount percentages.
7. Customer segmentation (New, Returning, Loyal).
8. Top products within each category.
9. Repeat buyers and subscription relationship.
10. Revenue contribution by age group.

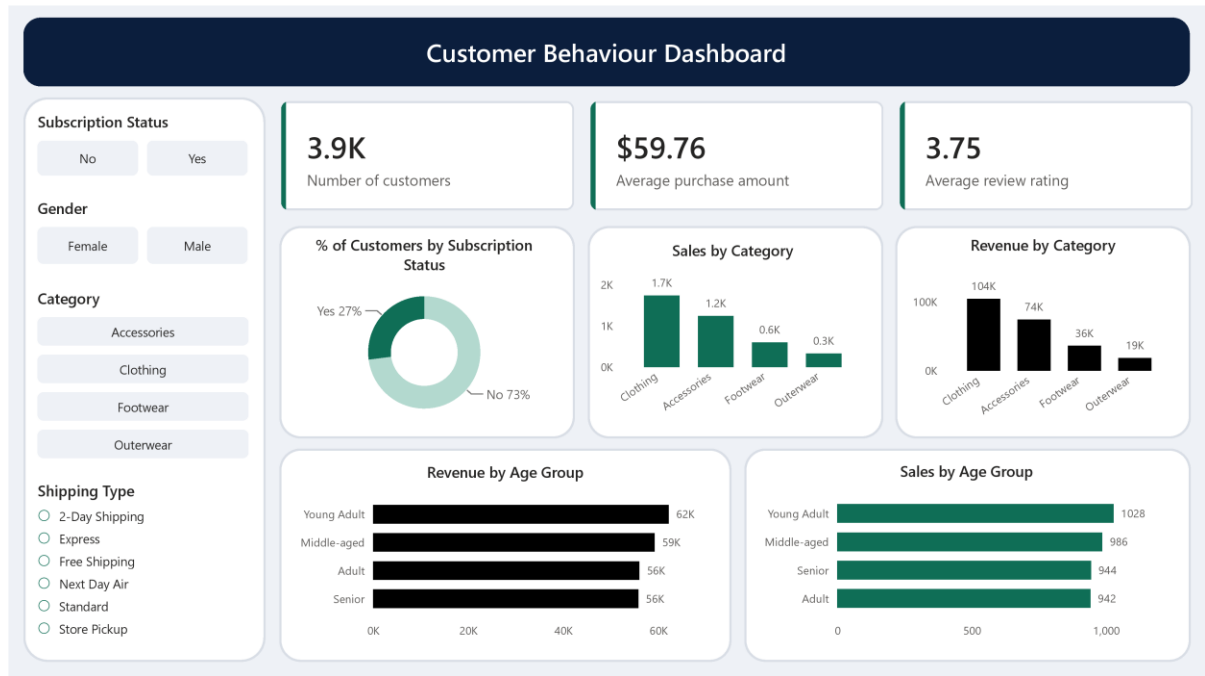
Note: Full SQL queries are available in the repository under `/sql/queries.sql`.

Dashboard Development

The Power BI dashboard was developed to provide stakeholders with an interactive view of customer behavior. Key dashboard components include:

- Total Customers KPI
- Average Purchase Amount KPI
- Average Review Rating KPI
- Subscription Analysis
- Sales by Category
- Revenue by Category

- Revenue by Age Group
- Sales by Age Group
- Interactive Filters



Key Insights

- Clothing category generated the highest revenue contribution.
- Young Adult customers contributed the highest revenue among age groups.
- Subscription behavior indicates opportunities for customer retention initiatives.
- Product categories show varying revenue performance that can guide inventory and marketing decisions.
- Customer segmentation enables targeted marketing and loyalty campaigns.

Business Recommendations

1. Increase Subscription Adoption

Offer exclusive benefits and personalized promotions to encourage subscriptions.

2. Strengthen Customer Loyalty Programs

Reward repeat customers and encourage movement toward the loyal customer segment.

3. Focus on High-Performing Categories

Allocate marketing resources toward top-performing categories.

4. Personalized Marketing Campaigns

Use demographic and behavioral insights for targeted promotions.

5. Data-Driven Decision Making

Continuously monitor customer behavior using dashboard analytics.

Project Deliverables

- Python Data Cleaning Notebook
- PostgreSQL Database Integration
- SQL Analysis Queries
- Interactive Power BI Dashboard
- Project Documentation
- GitHub Repository

Conclusion

This project demonstrates a complete data analytics workflow from raw data preparation to business intelligence reporting. By combining Python, PostgreSQL, SQL, and Power BI, meaningful customer insights were generated that can support strategic business decisions and improve customer engagement.